

MicroPulse

Load Intelligence — Coach Guide

The Internal:External coupling page — what works on Lite, what doesn't

FOR LITE CATAPULT TEAMS

1. Overview — what is Load Intelligence?

Load Intelligence (/coach/load-intelligence) is the page where MicroPulse stitches together every load signal it computes — both the external ones (what the GPS / IMU sensors recorded) and the internal ones (how hard the body worked). The page tells the Internal:External coupling story for your squad in a single scroll.

This guide is written specifically for Lite Catapult teams. The page renders the same 5 cards regardless of plan, but two of those cards depend on Pro / Premium parameters (B2-3 efforts, IMA Free Running) and will display partial or missing values on Lite. This guide tells you exactly which cards to trust on Lite, which ones to ignore, and what the Lite-equivalent insight is.

Companion docs

Two cards on this page have their own dedicated coach guides — read them alongside this one for the deeper detail. Training Monotony & Strain has a full guide. MD Comparison — HSR Exposure has a full guide. HSR Intelligence (a separate top-tab page on Lite) is the canonical hamstring-injury monitor and also has a full guide.

2. The five cards on the page

Load Intelligence renders cards in this order, top to bottom. The first column tells you what the card uses; the second tells you whether it works fully on Lite.

Card	What it shows	Lite tier verdict
GPS Load Intelligence	Per-player external GPS signals: NBS, decel burden, A:D coupling, HID%, residual decel, PL spike.	PARTIAL
Training Monotony & Strain (Foster 1998)	Per-player overtraining-risk monitor from sRPE × duration (or volume fallback).	FULL
MD Comparison — HSR Exposure (Buchheit 2018)	Per-player periodization helper — week's HSR vs personal match-day demand.	FULL
Mechanical Load Index (MLI)	Team-wide internal-load proxy from accel/decel + heart-rate work.	PARTIAL
Team Metabolic Summary	Team-wide metabolic load (di Prampero metabolic-power-derived volume).	PARTIAL

Bottom line for Lite teams

Two of the five cards (Foster Monotony & Strain, MD Comparison) work at full fidelity on Lite. They are the heart of your Load Intelligence workflow. The other three cards still render but show reduced or missing values because they need parameters Pro / Premium plans expose. Lean on the two FULL cards and treat the others as supplementary.

3. Card 1 — GPS Load Intelligence

This is the per-player external-signals table. It computes 6 signals from GPS / IMU data and flags players whose values are outside their personal baseline.

What it shows

- NBS — Normalised Body Stress score (0-100), an internal-load summary.
- Decel burden — count of high-intensity decelerations vs personal baseline.
- A:D Coupling — ratio of high-intensity accelerations to decelerations (eccentric vs concentric balance).
- HID% — share of high-intensity distance, tracked as a 7-day trend (Harper 2019 fatigue signal).
- Residual Decel — 3-day accumulated deceleration stress.
- PL Spike — Player Load deviation vs 7-day average.

Lite tier — what works and what doesn't

Signal	Lite status	Why
NBS	Works	Computed from total distance, HSR and player load — all available on Lite.
Decel burden	—	Requires deceleration B2-3 efforts (Pro / Premium parameter). Will show "—" on Lite.
A:D Coupling	—	Requires acceleration AND deceleration B2-3 efforts. Will show "—" on Lite.
HID% trend	Works	Computed from HSR vs total distance — both Lite metrics.
Residual Decel	—	Built on decel burden, so unavailable on Lite.
PL Spike	Works	Computed from player load — available on Lite.

How to read this card on Lite

Three of the six signals (NBS, HID% trend, PL Spike) work fully — these are real signals you should act on. The other three (decel burden, A:D coupling, residual decel) will show "—" because the underlying B2-3 efforts are not in your plan. Don't worry — Decel Intelligence on Pro plans does what those signals do; on Lite, you instead use HSR Intelligence (which is the published Lite-equivalent for hamstring injury monitoring).

Lite-equivalent for the missing decel signals

If you want decel-style injury insight on Lite, use **HSR Intelligence (top-tab on Lite)** — it gives per-player hamstring/soft-tissue injury risk from Malone 2017 + Buchheit 2014 metrics. Same evidence quality (BJSM-published), built on metrics every Catapult plan exposes.

4. Card 2 — Training Monotony & Strain (Foster 1998)

Lite verdict: WORKS FULLY

This card was designed before GPS existed. It needs only sRPE + duration (or total distance as a fallback). No B2-3 efforts. No premium parameters. This is the highest-quality training-load signal available to a Lite team.

What it shows

Per-player Monotony and Strain over the past 7 days. Monotony = mean ÷ SD of daily loads (high = no day-to-day variation = overtraining risk). Strain = weekly load × monotony (combined volume × variability).

Thresholds

Metric	SAFE	WATCH	DANGER / HIGH
Monotony	≤ 1.5	1.5 — 2.0	> 2.0
Strain	≤ 4500 AU	4500 — 6000	> 6000

Two variants — which is active?

- RPE × duration (green badge): canonical Foster method. Triggered when at least 5 distinct days have RPE entries logged in the squad's 7-day window.
- Volume (m) (blue badge): fallback when RPE compliance is too low. Uses GPS total distance ÷ 10 to scale to AU range. Weaker signal but better than nothing.

If you find your team consistently in the Volume variant, **fix RPE compliance** — even one player logging every day clears the threshold for the whole squad. The canonical RPE variant is meaningfully better.

For a full deep-dive on this card — math, scenarios, FAQ, limitations — see the dedicated Training Monotony & Strain coach guide.

5. Card 3 — MD Comparison — HSR Exposure (Buchheit 2018)

Lite verdict: WORKS FULLY

Built on high-speed running and (optionally) match minutes. Both available on every Catapult plan. This is the periodization tool you use to make sure each player has accumulated enough HSR before the next match.

What it shows

Per-player HSR this week vs personal match-day demand. The 95% rule (Buchheit 2018): players who reach ≥95% of their MD HSR by MD-1 enter the match prepared. Players under 80% across the cycle are at significantly elevated injury risk on match day.

Status flags

Status	% of MD	What you should do
UNDER	<80%	Add HSR to MD-1 (positional drills, possession on a larger pitch, short sprints), or reduce match minutes.
WATCH	80-95%	MD-1 should add 200-400 m of HSR to bring the player into the ready band.
READY	95-110%	Optimal. Keep the normal MD-1 plan (taper).
OVER	>110%	Make MD-1 lighter (recovery, technical) — player risks pre-match fatigue.

Two baseline strategies are picked automatically: "Match-derived" (preferred — uses median HSR from real matches the player started in) or "Synthetic" (pre-season fallback — uses top 10% of own training sessions). The strategy badge at the top of the card tells you which is active.

For a full deep-dive on this card — workflow, FAQ, edge cases — see the dedicated MD Comparison coach guide.

6. Card 4 — Mechanical Load Index (MLI)

Lite verdict: **PARTIAL**

MLI was designed for higher-tier data — its full calculation needs accel/decel B2-3 efforts and IMA bands. On Lite it can still produce a result from the metrics it does have (player load + heart rate where available), but the confidence indicator on each row will reflect the missing inputs.

What it shows

Team-wide MLI (0-100 scale) and the four sub-scores: DSS (deceleration), ASS (acceleration), CSS (change of direction) and GDS (general density). Plus residual MLI (3-day accumulated mechanical load) and per-player flags.

On Lite — what to do

- Read the team-wide aggregate but be aware confidence indicators may show "medium" or "low" because the canonical inputs are missing.
- DSS (deceleration sub-score) and CSS (change of direction sub-score) will show "—" or partial values — they need IMA bands.
- Treat MLI on Lite as a directional signal, not a precise number. For the same insight at full fidelity, the Lite path is Foster Monotony & Strain (above) — that one is your primary internal-load read.

7. Card 5 — Team Metabolic Summary

Lite verdict: **PARTIAL**

di Prampero metabolic power calculation requires Catapult to expose the `metabolic_power` column with high-resolution sampling. This is typically a Premium parameter — Lite plans rarely expose it. If your data doesn't include `metabolic_power` values, this card will show empty values or no rows.

What it shows

Team-wide metabolic load — total work done above the metabolic-power threshold, recovery time progression (#MPE recovery, Osgnach 2023), and Equivalent Distance Index (EDI).

On Lite — what to do

- Check the card briefly to see if any metabolic_power data is coming through. If yes, you can use the team-aggregate trends.
- If the card is empty, that confirms metabolic_power is not in your CSV / sync. This is normal on Lite.
- Lite-equivalent: total distance and HSR cover ~80% of what metabolic power would tell you for a typical session. Your primary external-load reads are HSR Intelligence (top-tab) and the GPS Data tab raw daily numbers.

8. Complete Lite-tier workflow

Here is how to use the Load Intelligence page end-to-end on a Lite Catapult plan, in a typical training week.

Step 1 — Daily morning load review (5 min)

- Open Load Intelligence. Scroll past Card 1 (GPS Load Intelligence) — note the NBS / HID% / PL Spike values for any flagged players, but ignore the "—" decel signals.
- Stop on Card 2 (Training Monotony & Strain). Read the table — sort is already by Strain descending. Players in DANGER need attention today; those in WATCH need a planned easy day soon.
- Glance at Card 3 (MD Comparison) only if you have a match in 1-3 days. Players in UNDER need HSR added to today's plan; OVER players need a lighter session.

Step 2 — Weekly match-week planning (15 min)

- Open MD Comparison (Card 3) early in the week (MD-3 or MD-2). Plan MD-1 sessions to bring all players into the READY band (95-110%).
- Cross-check Foster Monotony & Strain (Card 2). Are any players flagged DANGER going into the match? They need a recovery day in MD-1, not extra HSR. Make the trade-off: match readiness vs short-term fatigue.
- Use HSR Intelligence (separate top-tab) to see hamstring injury risk over the same period. RED on HSR Intelligence + DANGER on Foster = high-priority talk with the player.

Step 3 — Post-match calibration (10 min)

- After the match, the MD Comparison baseline auto-updates with the new match data point. Synthetic teams gradually become match-derived after 1-2 matches.
- Look at who was OVER going into the match — were they pre-fatigued and underperforming? Adjust their MD-1 plan for the next match.
- Foster Monotony & Strain often spikes after matches (high session load + low SD). Plan an easy day MD+1 to drop monotony fast.

9. Quick reference — Lite tier card behaviour

Card	Trustworthy on Lite?	Use it for
GPS Load Intelligence	Partial — only NBS, HID% and PL Spike are reliable	Spotting short-term player-load spikes and HID% fatigue trend. Ignore decel-related signals.
Training Monotony & Strain	Yes — fully reliable	Daily overtraining-risk triage. Your primary internal-load read.
MD Comparison	Yes — fully reliable	Match-week periodization. Make sure every player reaches ≥95% MD HSR by MD-1.
Mechanical Load Index	Partial — confidence indicators show what's missing	Directional team-aggregate trend. Don't rely on individual player MLI numbers in isolation.
Team Metabolic Summary	Often empty on Lite	Mostly skip on Lite. Use HSR Intelligence and GPS Data tab for external-load context instead.

10. Where Load Intelligence fits in your Lite toolkit

Load Intelligence is one of three pages in the Lite-tier load monitoring set. They complement each other — one page won't tell you everything.

Page	What it answers	When to open it
Load Intelligence (this page)	How are my players coping with this week's training? Are any in overtraining-risk territory?	Daily, 5 min. Triage screen.
HSR Intelligence (top-tab)	Which players are at elevated hamstring / soft-tissue injury risk based on chronic spike patterns and max-V exposure?	Daily after sessions sync. The injury-risk monitor.
Quadrant View Lite	How does each player sit on the Internal vs External load quadrant — overloaded, undertrained, or ramping cleanly?	Weekly. Big-picture periodization view.

11. What you'd unlock on Pro

If your club later upgrades to a Pro Catapult plan (and enables Acceleration & Deceleration B2-3 efforts in Reporting Parameters), the same Load Intelligence page automatically gains:

- Decel burden, A:D coupling, residual decel signals on Card 1 — not "—".
- Decel Intelligence (a separate top-tab page) — McBurnie 2022 4-dimension framework on B2-3 efforts. Higher-fidelity injury risk than HSR Intelligence.
- Higher-confidence MLI computation on Card 4 — full DSS sub-score.
- Sharp Cut + #MPE Recovery chips on Decel Intelligence (Premium tier).

The Lite-tier value is intentionally complete — every published evidence-based load monitoring algorithm has either a full version or a Lite-equivalent. The upgrade is about going from BJSM-published-equivalent to BJSM-published-original.

12. References

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